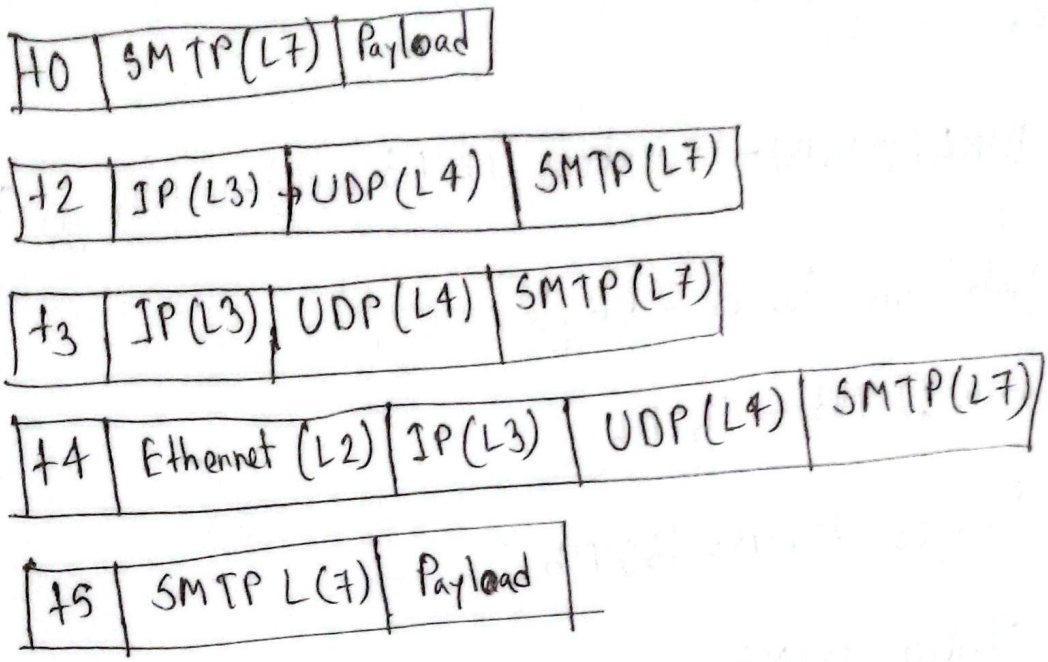
Q-1



Q-2

I. Transport Layer (Layer 4)

II. Physical Layer (Layer 1)

III. Transport Layer (Layer 4)

IV. Transport Layer (Layer 4)

V. [redacted] Application Layer (Layer 7).

Q-3

Given,

1 Base HTML (2 MB) + 18 Videos (100 MB each) + 7 Images (20 MB each)

1 Note on Units - 1 Byte = 8 bits

Base; 2 MB = 16 Mb

Images, 20 MB = 160 Mb

Video = 100 Mb

Bandwidth, R = 8 Mbps

RTT_{TCP} = 20 s

RTT_{HTTP} = 22 s

$T_{base} = \frac{16}{8} = 2s$

$T_{img} = \frac{160}{8} = 20s$

$T_{video} = \frac{100}{8} = 12.5s$

(i) Total round trip time = (20 + 22 + 2) sec

= 44 second.

(ii) Set up & Base HTML = 44 sec

18 videos = 18 x (22 + 12.5)

= 621 sec

7 Images = 7 x (22 + 20)

= 294 sec

Total = 44 + 621 + 294 = 959 s.

(iii) For Bob,

Total objects = 16

Time for 18 videos = 18 x (20 + 22 + 12.5)

= 18 x 54.5 s

= 981 s

Time for 7 images = 7 x (20 + 22 + 20)

= 7 x 62 s

= 434 s

N = 44 + 981 + 434 = 1459 s.

$N - M = (1459 - 959) \times 500s$

$24 \times TCP\ RTT = 24 \times 20s$

$= 480s$

$500s \neq 480s$

∴ The statement is false.

Q-4

Given,

TCP = 18s

HTTP RTT = 18s

Video speed = 90 Mbps

Other Speed = 45 Mbps

(1) Malcom waits for 1 setup + 1 Base Request + 24 object Request

Total waits = 26

∴ Total RTT = (26 × 18) = 468s.

(ii) ~~Roger Hotmail~~

10 videos + 14 other objects = 24 embedded objects

Base HTML file - [∵ 1 byte = 8 bits]

Size = (3 × 8) = 24 Mb

Speed = 45 Mbps

Time = $\frac{24}{45}$ = 0.533 s

Videos (10 files, 9 MB each) -

Size (1 video) = 9 × 8 = 72 Mb

Speed = 90 Mbps

Time (1 video) = $\frac{72}{90}$ = 0.8 s

∴ For 10 videos = (10 × 0.8) = 8.0 s

Other objects (14 files, 2 MB each)

Size (1 obj) = 2 × 8 = 16 Mb

Speed = 45 Mbps

Time (1 obj) = $\frac{16}{45}$ = 0.356 s

for 14 obj = 14 × 0.356 = 4.98 s

Total transmission time = (0.533 + 8.0 + 4.98) s

= 13.51 s

∴ Total time = (468 + 13.51) s

= 481.51 s

(14) Malcolm's Connection - 1 setup

Reese's Connection - 25 setup

Reese performs 24 more setups than Malcolm.

Extra Time = Extra Setups × Time per setups

= (24 × 18) sec

= 432 sec

∴ Reese's total time increased by 432 seconds compared to Malcolm.

∴ The TCP Connection Establishment time is the most affected part cause this is the specific steps Reese repeats 24 extra times.

Q-5

(i) HEAD as this method retrieves the headers without transferring the body, saving bandwidth.

(ii) POST as it is used to submit data to the server to be processed.

(iii) PUT as it is used to upload a resource to a specific URL, replacing any existing version at that location.

Q-6

(i) forward the request as one cannot cache a login request. It contains a username and password that the real server must check. If the proxy just send back a saved response, the user wouldn't actually be logged in.

(W) Delete the old image. The proxy saw the PUT request go through, so it knows the image has been updated. It must delete on mark it's old copy so that the next person who asks for it gets the new version, not the old one.